GE2340 Data Analysis and Baseline Results

Technical evidence | 3 November 2026 | Regression method verified against submitted Phase 1

This report documents the dataset, exploratory analysis, experimental settings and baseline results for the team report. Data preparation for continuous prediction is complete, and Section 6 defines the classification target. The regression method matches the submitted Phase 1 proposal.

1 Dataset and data quality

- Source: *Lifestyle Factors and Their Impact on Students*, Charlotte Bennett1234, Kaggle version 3. The original file contains 2,000 rows and 9 columns.
- The outcome, Grades, retains its original units and ranges from 5.60–10.00 in the full dataset.
- Predictors: Study_Hours_Per_Day, Extracurricular_Hours_Per_Day, Sleep_Hours_Per_Day, Social_Hours_Per_Day and Physical_Activity_Hours_Per_Day. Each measures hours per day.
- Student_ID identifies records only. Gender and Stress_Level are excluded to match the five-input Phase 1 plan.

Quality check	Result
Missing cells in the original file	0
Exact duplicate rows / duplicate student IDs	0 / 0
Rows removed / imputed	0 / 0
Clean observations retained	2,000
Daily durations	All five sum to 24 within floating-point tolerance

We checked the schema, numeric values for finiteness, ID uniqueness and whether durations fell within 0–24 hours. All observations were retained without imputation. Training-set interquartile range (IQR) flags identify potential outliers.

Evidence: quality_audit.json; student_lifestyle_dataset..csv; run_pipeline.py.

2 Data preparation and descriptive statistics

- With a fixed random seed of 2340, we reserved 20% of observations for testing. Of the remaining 80%, 25% were assigned to validation. This left 1,200 training, 400 validation and 400 test observations.
- Exploratory plots and correlations use only the training set. StandardScaler was fitted to the training predictors, then applied unchanged to the validation and test sets. Grades was not standardized.
- The shared data exports retain all five predictors. The linear baseline alone uses four, with physical activity as the reference activity.

Training variable	Mean	Sample SD	Min	Max
Study	7.494	1.442	5.000	10.000
Extracurricular	1.984	1.136	0.000	4.000
Sleep	7.496	1.459	5.000	10.000
Social	2.720	1.702	0.000	6.000
Physical activity	4.306	2.475	0.000	13.000
Grades	7.777	0.763	5.600	9.820

Training sample n = 1,200. Durations are measured in hours/day, and Grades retains its original units. Sample SD uses n−1 as the denominator. StandardScaler uses the population convention, so the SD values above differ from its scaling parameters.

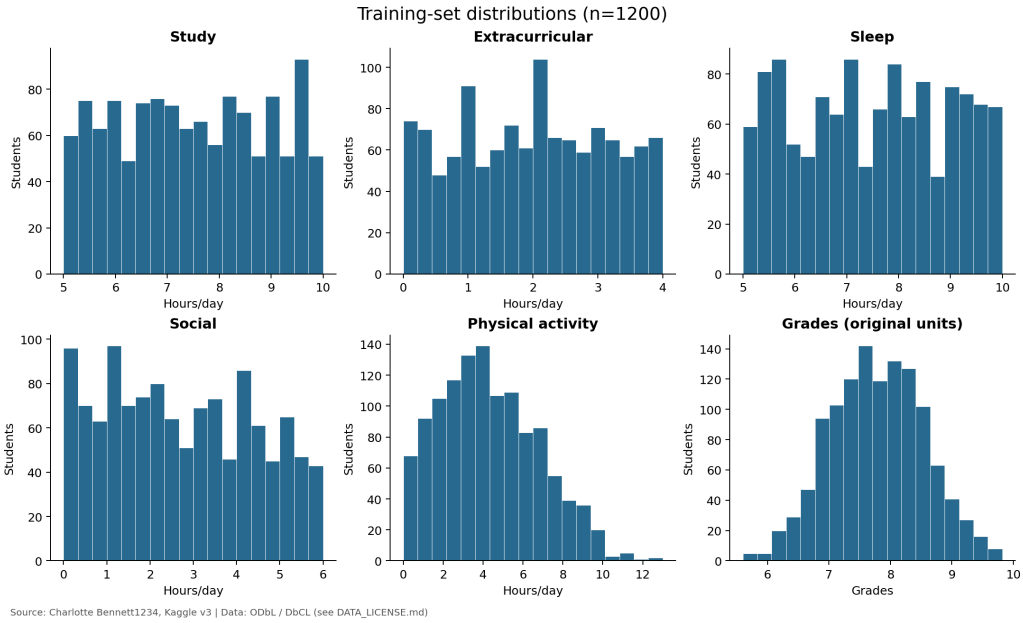


Figure 1. Training distributions, showing the observed ranges and shapes.

3 Exploratory relationships and interpretation

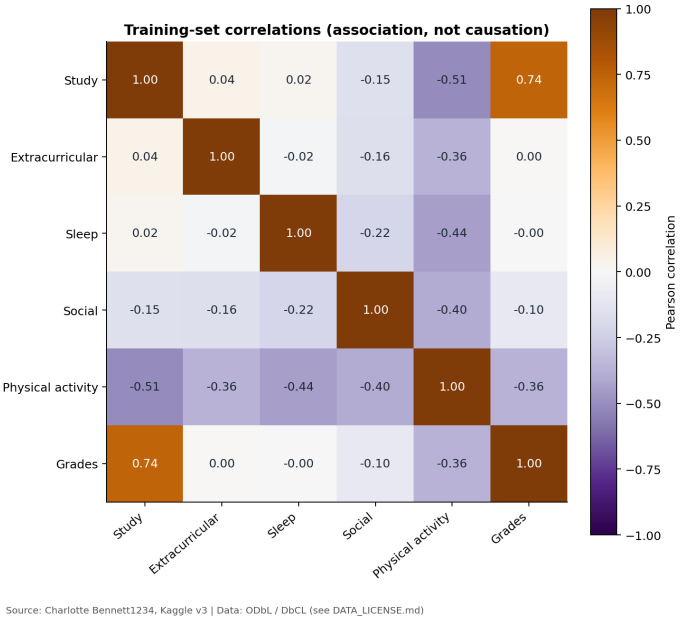


Figure 2. Training Pearson correlations, n = 1,200. Source attribution appears on the figure.

Predictor	Correlation with Grades
Study	0.7391
Physical activity	−0.3614
Social	−0.1009
Extracurricular	0.0024
Sleep	−0.0015

- Among the measured predictors, study time has the strongest positive linear association with Grades. Sleep and extracurricular time have near-zero marginal correlations with Grades.
- Because the activities total 24 hours, increasing one requires reducing one or more others. Study and physical activity have a correlation of −0.5115.

Evidence: train_correlations.csv. The scatter-plot figure in the original package provides a complementary view of these same training observations.

4 Linear regression and baseline evaluation

- The mean baseline predicts 7.777125 for every observation. This is the mean Grades value among the 1,200 training observations.
- We fitted the linear model with scikit-learn LinearRegression, using an intercept and ordinary least squares (OLS). The squared-error objective matches the mean squared error (MSE) objective in the submitted Phase 1 proposal. The model uses unstandardized training hours and returns raw predictions in the original Grades units.
- The five durations sum to 24, and their centered training matrix has rank 4. Setting physical activity as the reference removes coefficient redundancy. Its information is retained through the total-time constraint.

Split	Model	MAE	RMSE	R^2
train	linear regression	0.4067	0.5131	0.5474
train	mean baseline	0.6234	0.7626	0.0000
validation	linear regression	0.4104	0.5183	0.5147
validation	mean baseline	0.6049	0.7442	-0.0005
test	linear regression	0.3725	0.4731	0.5403
test	mean baseline	0.5659	0.6994	-0.0044

MAE measures mean absolute prediction error, and RMSE measures root mean squared error. Both are expressed in the original Grades units, with lower values indicating smaller errors. R^2 compares squared prediction error against predicting the evaluation set's own mean. Higher R^2 is better; negative values are possible.

- On the test set, linear regression had an MAE of 0.372452, compared with 0.565948 for the mean baseline. This represents a reduction of about 34.2%. Test R^2 was 0.540324.
- Training and validation MAE were 0.406691 and 0.410385. The model produced similar prediction errors on these two sets.

Phase1_Proposal.pdf, page 1, specifies mean squared error and obtains the best-fit parameters by setting the derivative to zero. The OLS method used here follows that objective.

5 Validation diagnostics and experiment status

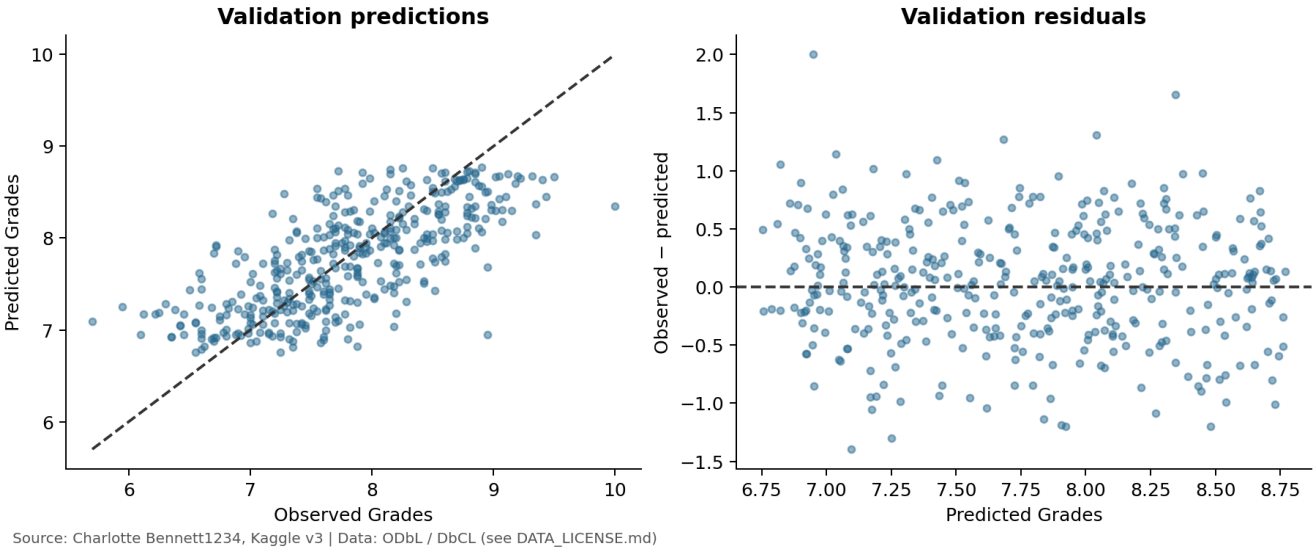


Figure 3. Validation diagnostics, n = 400. Residuals are observed minus predicted Grades.

- The dataset's original collection method and real or synthetic provenance remain unverified.
- The OLS model and mean baseline have been evaluated on the fixed data split.
- Section 6 defines the classification target. Softmax results, including classification accuracy and Macro-F1, are pending. The neural-network experiment is optional.

Evidence and source references

Primary dataset: Charlotte Bennett1234. Lifestyle Factors and Their Impact on Students. Kaggle, version 3.

Dataset license terms: Open Database License 1.0 and Database Contents License 1.0. Figures and derived tables retain source attribution.

Reproducible evidence: metrics.csv; regression_model.json; train_descriptive_statistics.csv; train_correlations.csv; split_manifest.csv; preprocessor.json; run_pipeline.py. Raw source metadata and the executable pipeline are included in the data package.

Project status

- The grade bands in Section 6 are fixed.
- The regression baseline uses the squared-error OLS objective and four predictors under the fixed 24-hour constraint.
- Softmax results remain to be integrated into the team report. Include neural-network results if the optional experiment is completed.

6 Project-defined classification target

The project defines relative grade bands from the training distribution. F denotes the lowest relative group, not an official failure category.

The training 20th, 40th, 60th and 80th percentiles are 7.10, 7.55, 8.00 and 8.45. We applied these fixed boundaries unchanged to validation and test observations. Equal scores share a label, so group sizes may differ.

Band	Grades interval	Train	Validation	Test
F	Grades < 7.10	238	69	59
D	7.10 ≤ Grades < 7.55	236	79	78
C	7.55 ≤ Grades < 8.00	234	99	104
B	8.00 ≤ Grades < 8.45	249	62	75
A	Grades ≥ 8.45	243	91	84

The intervals above define the classification labels.

The boundaries were computed from training data. The v1 test regression results had already been inspected when classification was added.

The shared data package contains the split, preprocessing parameters and classification labels.

Majority-class baseline

The majority-class baseline always predicts B, the most frequent training class (249 of 1,200). It provides a reference for evaluating subsequent classifiers.

Split	Accuracy	Macro-F1	Balanced accuracy
train	20.75%	0.06874	0.20
validation	15.50%	0.05368	0.20
test	18.75%	0.06316	0.20

Macro-F1 gives equal weight to each class. Balanced accuracy is mean class recall. Evidence: class_counts.csv and classification_metrics.csv in v2.